3	(a)	A solenoid shown in Figure 3.1 has length of 10 cm and 300 turns of wire. The current in the solenoid is 2.5 A when it is connected to a power supply.		

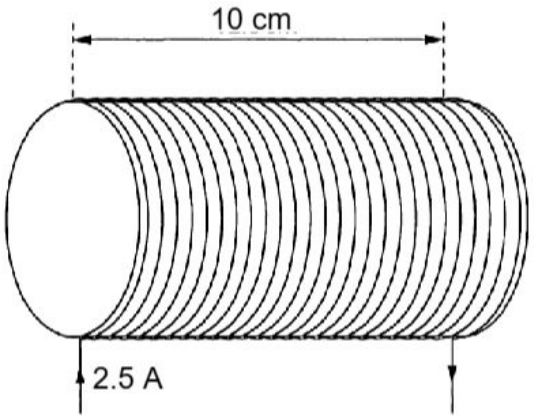


Figure 3.1

		(i)	Calculate the magnetic flux density inside the solenoid due to current of 2.5 A.	
			magnetic flux density = ...T	[2]
		L1	$B = \mu_o nI = (4\pi \times 10^{-7})(300 / 0.10)(2.5)$	M1
			$= 9.4248 \times 10^{-3}$	
			$= \mathbf{9.42 \times 10^{-3}\ T}$	A1
		(ii)	Inside the dashed space provided in Figure 3.2, sketch the magnetic field pattern due to the current in the solenoid when 2.5 A of current is flowing in it as shown. Indicate the polarity of the magnetic field with **N** as the north pole and **S** as the south pole.	

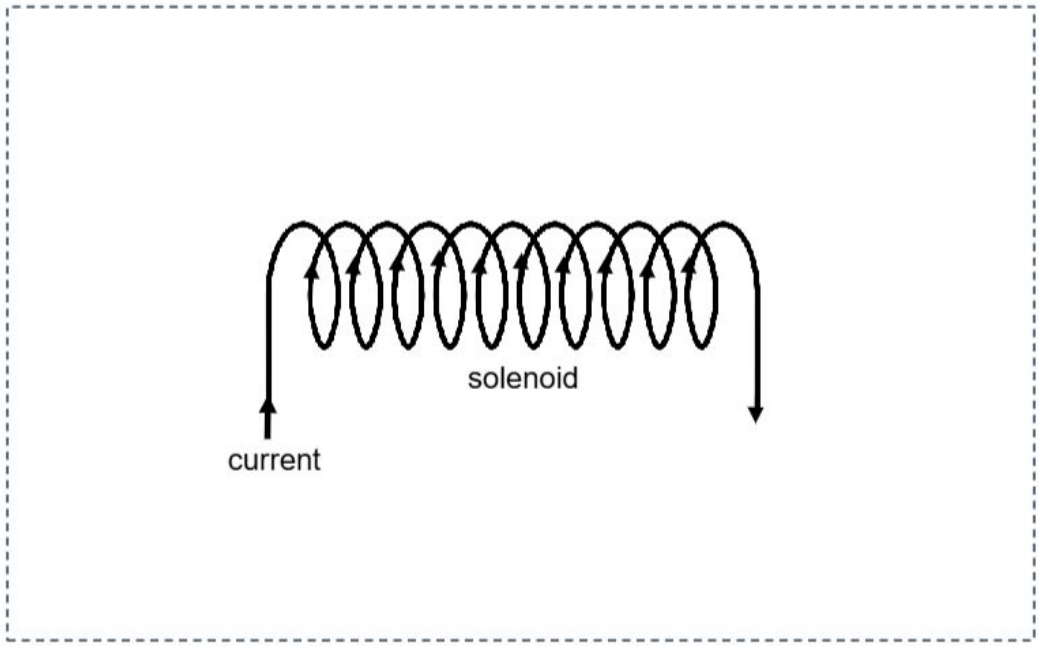
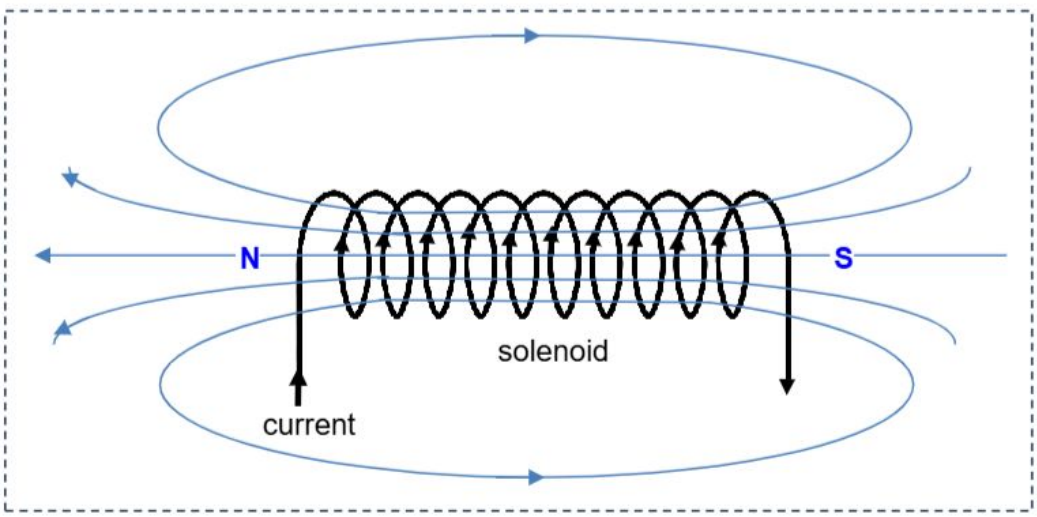


Figure 3.2 [2]

L2 **Solution:**



B1:

- **Correct pattern (correct shape; straight parallel equally spaced fields lines in the centre of the solenoid, diverge towards the ends of the solenoid; separation of field lines closer inside the solenoid compared to outside of it.)**
- **At least 5 lines drawn to show the trend/pattern.**
- **Field lines should not touch one another.**

B1:

- **Correct direction of field lines.**
- **Correct polarity (N and S) indicated.**

(b) A pair of concentric coils made of an inner coil Y placed within a solenoid is shown in Figure 3.3. The solenoid has cross-sectional area of 3.0×10^{-2} m^2 and is made of 2500 turns of wire. The solenoid is connected to a variable power supply by the terminals CD.

Coil Y has 500 turns of wire, a cross-sectional area of 7.5×10^{-3} m^2 and resistance of 3.0 Ω. Coil Y is connected to a resistor R of 15 Ω.

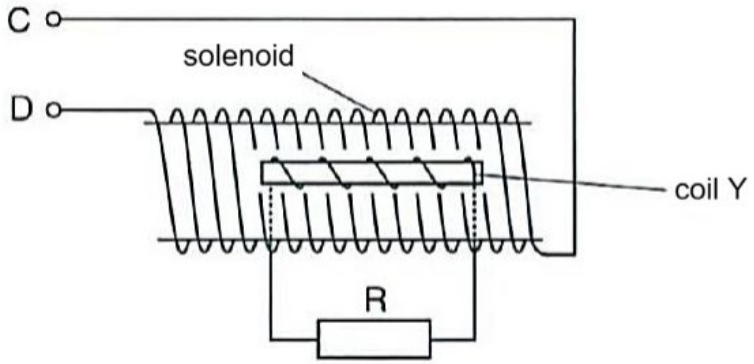


Figure 3.3

The current in the solenoid is then varied causing the magnetic flux density *B* in the solenoid to vary with time *t* as shown in Figure 3.4.

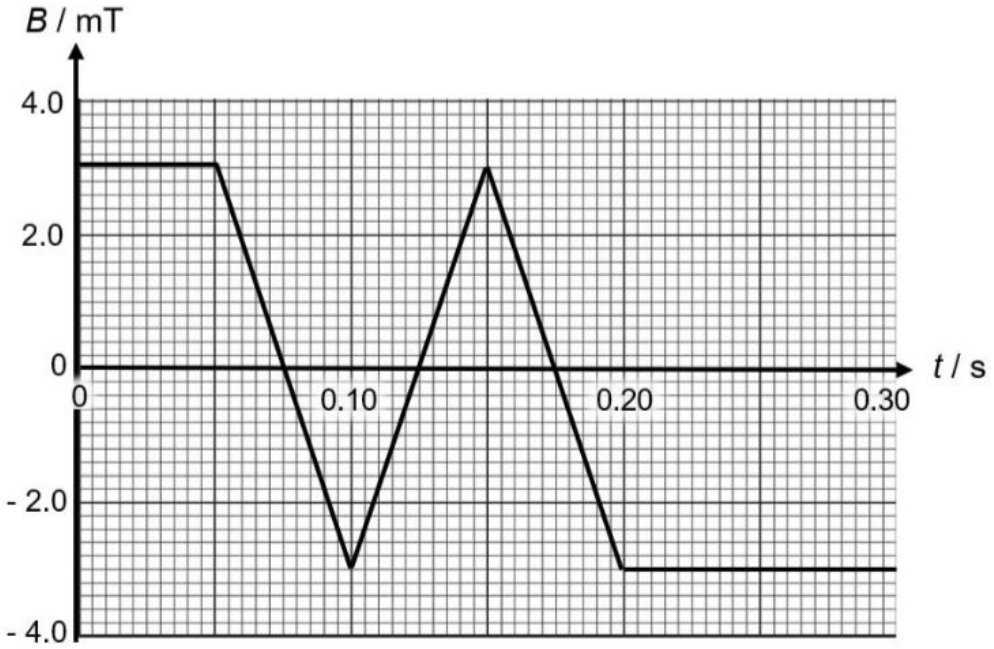


Figure 3.4

		(i)	Calculate the magnitude of the maximum current in R.

current = ...A [3]

L2 Magnitude of **maximum** Induced EMF

$$E_Y = -\frac{d\Phi_Y}{dt} = -\frac{d\left[N_Y B_S A_Y\right]}{dt} = -N_Y A_Y \frac{dB_S}{dt}$$

$$= -(500)\left(7.5\times10^{-3}\right)\left[\frac{\left(3.0-(-3.0)\right)\times10^{-3}}{0.050}\right]$$ ***(use the maximum gradient)*** **M1**

$$= -0.45 \text{ V}$$

$$|E_Y| = 0.45 \text{ V}$$

Magnitude of **maximum** current **M1**

$$I_{ind} = \frac{E_{ind}}{R} = \frac{0.45}{3.0+15}$$ **A1**

$$= 0.025 \text{ A}$$

		(ii)	On Figure 3.5, use your answer in **(b)(i)** to sketch a graph to show the variation with time of the current I in resistor R for time t = 0 to t = 0.30 s.

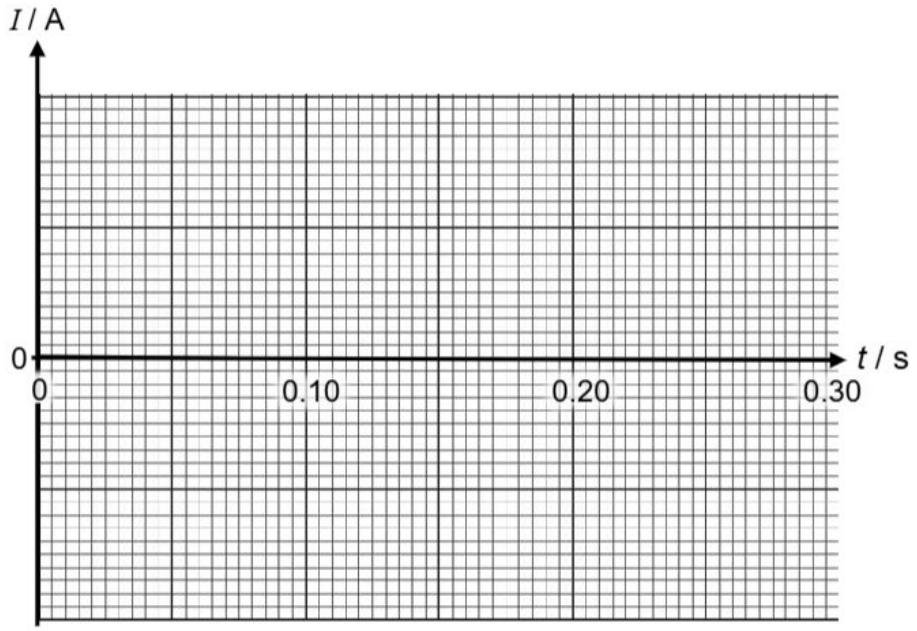
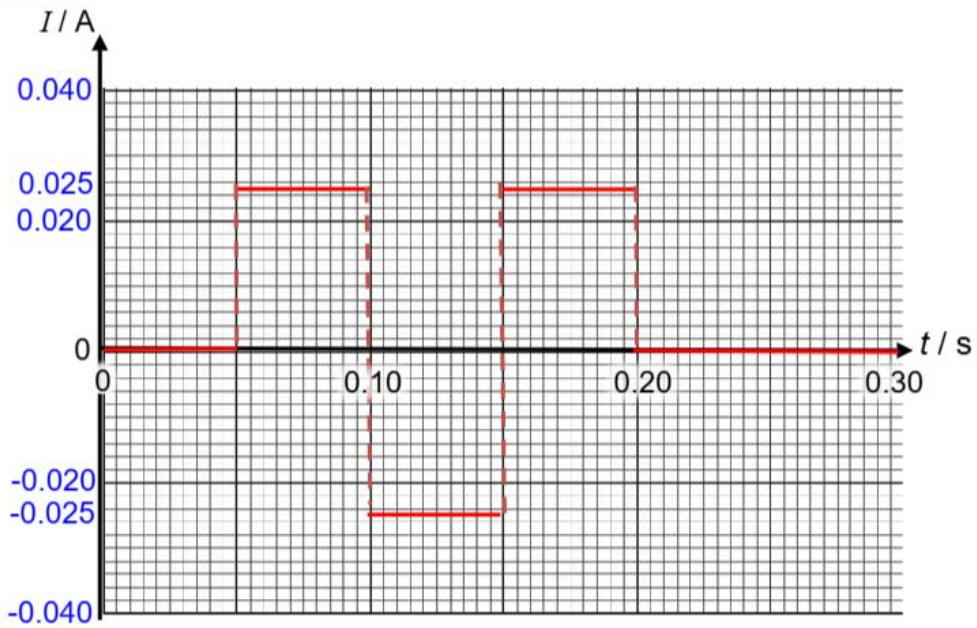


Figure 3.5 [2]

L2 **Solution:**



B1:

- **Where I = 0 from t = 0 to t = 0.05 s, and, t = 2.0 s to t = 3.0s.**

B1:

- **Maximum value of I from (b)(ii) is plotted accurately to ½ the smallest division. [Allow ECF.]**
- **Where $\frac{dB}{dt}$ is constant and positive, I is constant and negative, and**
- **Where $\frac{dB}{dt}$ is constant and negative, I is constant and positive.**
- **Good scale for y-axis: 1:1, 1:2, 1:4, 1:5 and graph covers at least half the graph grid vertically.**

[Total: 9]